Two HRS Core cognitive classification methods informed by HRS/HCAP

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Abstract

- **Importance:** population cognitive classification in HRS Core; dementia and MCI surveillance; limited assessment battery
- **Objective:** describe two HRS Core methods calibrated to HRS/HCAP (core algorithm, PMM); compare with the Manly-Jones (HCAP) algorithm, consensus, Langa-Weir, and Hudomiet
- **Design:** 2016 HRS Core classification in HRS/HCAP participants; consensus-panel secondary comparator
    - Study sample: HRS/HCAP 2016 participants, N = 3,496
    - [Provenance: R/MS_MAIN/MS_001_analysis_cohort.R]
- **Methods:** Core algorithm; PMM
    - Primary comparator: Manly-Jones algorithm
    - Secondary comparators: (1) HCAP consensus; (2) Langa-Weir; (3) Hudomiet classification
    - Statistical analysis: dementia versus non-dementia kappa; 3-class quadratic weighted kappa (Normal, MCI, Dementia); all statistics other than frequencies use complex sampling weights
    - [Provenance: R/MS_MAIN/MS_010_results_objects.R]
- **Results:**
    - Dementia/non-dementia kappa
        - Core algorithm versus Manly-Jones algorithm: 0.337
        - PMM versus Manly-Jones algorithm: 0.302
    - 3-class kappa
        - Core algorithm versus Manly-Jones algorithm: 0.501
        - PMM versus Manly-Jones algorithm: 0.37
    - [Provenance: R_objects/A7_100_hcap_tables.rds; R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]
- **Conclusions and relevance:** Two core classification approaches; agreement evidence rather than clinical validation; uncertainty in distinguishing MCI from normal cognition and dementia; limited precision of comparisons with consensus diagnoses because the consensus-panel subsample is small
    - [Provenance: analysis design; inference]

Key terms

- **Manly-Jones (HCAP) algorithm**: c.f. Manly, Jones et al. (2022)
- **HRS core algorithm**: Manly-Jones algorithm adapted to HRS core data

- **PMM**: Profile mixture model using measures collected in the HRS Core, calibrated to the Manly-Jones algorithm and scored in the HRS/HCAP sample

Introduction

- Population cognitive classification from brief HRS Core assessments
    - Prevalence estimation; etiologic research; trial sampling; policy surveillance
    - Core assessment constraints; incomplete cognitive testing; proxy/informant information
- Manly-Jones (HCAP) algorithm as primary comparator
    - HCAP neuropsychological assessment; actuarial Normal/MCI/Dementia classification
    - Manly-Jones algorithm; primary HCAP-based comparator
    - Manly et al. (2022); Farron et al. (2025)
    - [Provenance: R/A7_100-comparison_of_diagnoses.R; existing manuscript materials]
- Study aim
    - HRS core algorithm and PMM; Manly-Jones algorithm primary comparator
    - Secondary comparators: HCAP consensus; Langa-Weir; Hudomiet
    - Classification margins; 3-class agreement; dementia/non-dementia agreement

Methods

Design and analytic samples

- Study sample: HRS/HCAP 2016 participants, N = 3,496
    - All HCAP participants; no Core cognitive-test requirement
    - Participants without Core cognitive testing classified from informant data (Jorm IQCODE): core algorithm Jorm fallback; PMM informant model
    - Agreement-statistic samples: core algorithm N = 3,496; PMM N = 3,496
    - [Provenance: R/MS_MAIN/MS_001_analysis_cohort.R; R_objects/A7_005_hrs16_merged.rds; R/A7_100-comparison_of_diagnoses.R; R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]
- Consensus panel subset
    - Validation-subsample design; inverse-probability weighting to account for selection into the consensus panel
    - [Provenance: R/A7_100-comparison_of_diagnoses.R; R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]

HRS Core classification approaches

- Two Core classification approaches
    - Core algorithm
        - Survey-weighted CFA; demographically normed cognitive score; cognitive/self-report/informant rules

        - [Provenance: R/A7_050-mplus_model.R; R/A7_055-norm_factor_score.R; R/A7_075-implement_algorithm.R]
    - PMM
        - HRS/HCAP-calibrated probabilistic Normal/MCI/Dementia classification; combined cognition and informant models (i.e., two PMM models, one trained using cognition data, the other trained using informant data (Jorm IQCODE)).
        - *Limitation*: The HRS/HCAP IQCODE look back window is the past 10 years, the Core IQCODE has a look back window of 2 years.
        - [Provenance: R/PMM_102_calibration_model.R; R/PMM_103_informant_model.R; R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]

- Single-factor Core cognition model
    - Not enough items to separately score 5 domains matching Manly et al. (2022)
    - Initial 8-indicator, complex-survey model fit evaluated with WLSMV
    - Factor scores estimated with MLR, a probit link, and EAP scoring after fixing item parameters to the MLR model estimates
    - Factor-score normalization; age, sex, race/ethnicity, education covariates
    - [Provenance: R/A7_050-mplus_model.R; mplus_output/A7/model_1a_fixed/model_1a_fixed.inp; R/A7_055-norm_factor_score.R]
- Rule-based classification
    - Cognitive thresholds; self-reported concern; JORM for missing cognitive assessments
    - Normal, MCI, Dementia output
    - More detail in **Appendix 1**
    - [Provenance: R/A7_075-implement_algorithm.R]

- HRS/HCAP calibration
    - Known-class Manly-Jones algorithm classification (training data; Muthén, 2002)
    - Same inputs as core algorithm: 8 Core cognitive tests; self-rated memory (D102); Jorm IQCODE
        - Not included: ADL/IADL items (reason for a limitation not asked; physical causes); immediate word recall (little beyond delayed recall; Jones et al., 2024)
    - Cognition model, N = 3,283 (at least 1 test score or D102)

        - General cognitive factor (GCP) on the 8 tests; GCP mean and variance by class (Normal: mean 0, variance 1)
        - Loadings, intercepts, thresholds, residual variances equal across classes; D102 thresholds by class
        - Test regressions on 24 sociodemographic terms fixed at robust norms group values
    - Informant model, N = 213 (IQCODE, no test score or D102)
        - Jorm IQCODE only; mean by class
    - Calibration parameters fixed for scoring; scored in the same HRS/HCAP sample used for calibration (in-sample classification); HRS Core 2016-2022 scoring in Analysis A3_2 (not reported here)
    - [Provenance: R/PMM_100_norming_npb.R; R/PMM_102_calibration_model.R; R/PMM_103_informant_model.R; R/PMM_090_model_registry.R]
- Choice of a general factor
    - Tests correlate within class (robust norms group, after adjustment); model without GCP sets within-class correlation to 0, treats correlated tests as independent evidence
    - GCP model: better fit, fewer parameters than no-factor model
    - Class-specific test intercepts and thresholds: 10-fold cross-validation; higher kappa, lower log score; not adopted
    - [Values not in a current results object; Provenance: References/SESSION_LOGS.md, 2026-10-22; R/Archive_PMM_2026-10-22/]
- Class proportions (priors)
    - Class-specific parameters estimated with HCAP16WGTR; MLR estimation, Bayes-rule posterior classification (plug-in rule; McLachlan, 1992)
    - Class proportions fixed at the HCAP16WGTR-weighted Manly-Jones algorithm distribution, computed outside Mplus; Mplus KNOWNCLASS estimates class proportions from unweighted counts
        - Cognition model: Normal 72.0%, MCI 20.7%, Dementia 7.3%
        - Informant model: Normal 31.7%, MCI 31.6%, Dementia 36.7%
    - [Provenance: R/PMM_101_mplus_function.R; R/PMM_102_calibration_model.R; R/PMM_103_informant_model.R; R/Archive_PMM_2026-10-22/PMM_104_weight_check.R; scoring outputs in R/PMM_090_model_registry.R]
- Combined classification
    - Cognition-model probabilities; informant-model probabilities when IQCODE available

    - Probabilistic Normal/MCI/Dementia membership; replicated probability-weighted comparisons
    - More detail in **Appendix 2**
    - [Provenance: R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]

Comparison classifications and estimands

- Primary comparator: Manly-Jones algorithm
    - HCAP population weights: HCAP16WGTR
    - [Provenance: R/A7_100-comparison_of_diagnoses.R]
- Secondary comparators
    - HCAP consensus diagnosis: normalized HCAP16WGTR x samplingP^-1 weights
    - Langa-Weir; Hudomiet classification
    - [Provenance: R/A0_017-recode_classification.R; R/A7_100-comparison_of_diagnoses.R]
- Agreement estimands
    - 3-class quadratic weighted kappa
    - Dementia versus non-dementia kappa
    - Marginal Normal/MCI/Dementia distributions
    - [Provenance: R/MS_MAIN/MS_010_results_objects.R]

Results

Analytic samples and classification margins

- Table 1: HRS/HCAP study sample characteristics
    - Dynamic counts and descriptive statistics
    - [Provenance: R/MS_MAIN/MS_Tab_Fig_Apndx-030-Table-1.qmd]
- Table 2: weighted classification margins
    - Manly-Jones algorithm; core algorithm; PMM; Langa-Weir; Hudomiet
    - HCAP16WGTR-weighted margins
        - Manly-Jones algorithm: Normal 69.4%, MCI 21.4%, Dementia 9.2%
        - Core algorithm: Normal 68.9%, MCI 22.0%, Dementia 9.1%
        - PMM: Normal 85.5%, MCI 8.6%, Dementia 5.8%
    - [Provenance: R/MS_MAIN/MS_010_results_objects.R]
- Figure 1: Core algorithm
    - Classification rules and participant counts
    - [Provenance: R/MS_MAIN/MS_Tab_Fig_Apndx-020-Figure-1.qmd]

- Figure 2: PMM class-probability distributions
    - Largest MCI probability 0.51; 2 of 3,283 above 0.50 (cognition model)
    - Ceiling from GCP class distributions and priors: 0.60 (GCP known); 0.46 (within-class omega 0.50)
    - Most likely class MCI: 313; mean MCI probability 0.46
    - [Provenance: R/PMM_114_mci_ceiling.R]
    - Probability separation; uncertainty near class boundaries
    - [Provenance: R/PMM_112_Margins.R]

Agreement with Manly-Jones algorithm

- Core algorithm versus Manly-Jones algorithm: 3-class kappa 0.501; binary kappa 0.337
    - [Provenance: R_objects/A7_100_hcap_tables.rds]
- PMM versus Manly-Jones algorithm: 3-class kappa 0.37; binary kappa 0.302
    - [Provenance: R/PMM_110_Comparison_of_Consensus_Langa_Weir.R]
- Table 3: pairwise agreement, comparators, weight definitions
    - [Provenance: R/MS_MAIN/MS_010_results_objects.R]

Agreement with consensus and established HRS classifications

- Core algorithm, PMM, Langa-Weir, Hudomiet
    - Separate HCAP population and consensus-subsample estimands
    - Weight-dependent interpretation
    - [Provenance: R/MS_MAIN/MS_010_results_objects.R]
- Table 4: A7 pairwise three-class agreement matrix
    - [Provenance: R_objects/A7_100_hcap_tables.rds]

Discussion

- Two HRS Core classification approaches
    - Rule-based core algorithm; probabilistic PMM classification
    - Same HRS/HCAP learning target; distinct measurement and classification assumptions
    - [Provenance:analysis methods]
- Primary comparison
    - Manly-Jones algorithm; agreement and marginal-distribution evidence
    - [Provenance: Tables 2-4]
- Secondary comparisons
    - Consensus: small validation subset; inverse-probability weights

    - Langa-Weir and Hudomiet: established HRS comparators; different development targets
    - [Provenance: R/A7_100-comparison_of_diagnoses.R]
- Interpretation limits
    - Classification agreement; no clinical diagnostic validation outside HCAP consensus subset
    - MCI boundary; model-dependent and threshold-dependent assignments
    - PMM agreement with the Manly-Jones algorithm is in-sample (apparent error rate; McLachlan, 1992); expected to be optimistic relative to application in new samples
    - MCI: Core battery, 1 dimension measured with error, no informant report for tested respondents; PMM MCI probabilities rarely above 0.50 (Results, Figure 2)
    - [Provenance: analysis design; inference; R/PMM_102_calibration_model.R; R/PMM_114_mci_ceiling.R]

Two HRS Core cognitive classification methods informed by HRS/HCAP: Figures, tables, and appendices

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Table 1. Participant Characteristics

Characteristic	HRS/HCAP 2016 participants (N = 3,496)
Sex (N (%))	
Female	2,095 (59.9%)
Male	1,401 (40.1%)
Age in years (Mean (SD))	76.0 (7.6)
Race (N (%))	
White	2,755 (78.8%)
Black	563 (16.1%)
All other groups	177 (5.1%)
Not collected	1 (0.0%)
Hispanic origin (N (%))	
Not Hispanic	3,111 (89.0%)
Hispanic	383 (11.0%)
Not obtained	2 (0.1%)
Years of education (Mean (SD))	12.7 (3.2)
HRS/HCAP Classification (N (%))	
Normal	2,313 (66.2%)
Mild cognitive impairment	808 (23.1%)
Dementia	375 (10.7%)

Table 1. Participant Characteristics

Characteristic	HRS/HCAP 2016 participants (N = 3,496)
Langa-Weir Classification (N (%))	
Normal	2,432 (69.6%)
Cognitively impaired, not demented	731 (20.9%)
Dementia	333 (9.5%)
Jorm IQCODE score from Core (Mean (SD))	3.8 (0.8)
Jorm IQCODE score from Core (missing)	3,283

Source: R/MS_MAIN/MS_001_analysis_cohort.R; R_objects/A7_005_hrs16_merged.rds.

Jorm IQCODE was collected in the HRS Core interview, not HCAP, and only from participants who did not complete Core cognitive testing.

Notes: The study sample includes all 3,496 HRS/HCAP 2016 participants, including participants without Core cognitive testing who were classified from informant (Jorm IQCODE) data. Race and ethnicity data were gathered via self-report at the time of first interview and are considered to be markers of exposure to evolving systems of racism, not as a proxy for genetic variation or any other biological variable. Race was self-selected by participants at the time of the first interview from a list of options defined by the 2000 US Census criteria. Other race includes a pooled group of participants who identified as American Indian or Alaska Native, Asian, Native Hawaiian or Pacific Islander, or another self-described race, consolidated due to small sample sizes and risk of identification.

Table 2. HCAP classification margins by method

	Normal		MCI		Dementia	
Method	N	Weighted %	N	Weighted %	N	Weighted %
Manly-Jones algorithm	2,313	69.4%	808	21.4%	375	9.2%
Core algorithm	2,358	68.9%	794	22.0%	344	9.1%
PMM	2,934	85.5%	330	8.6%	232	5.8%
Secondary comparators						
Langa-Weir	2,432	74.9%	731	17.3%	333	7.9%
Hudomiet	2,326	71.1%	865	21.0%	286	7.2%

Note. N values are unweighted HRS/HCAP participant counts. Percentages use HCAP16WGTR; PMM counts and percentages use the most likely class from the combined cognition and informant models (non-replicated).

References. Manly, J. J., Jones, R. N., Langa, K. M., Ryan, L. H., Levine, D. A., McCammon, R., Heeringa, S. G., & Weir, D. (2022). Estimating the prevalence of dementia and mild cognitive impairment in the US: The 2016 Health and Retirement Study Harmonized Cognitive Assessment Protocol Project. JAMA Neurology, 79(12), 1242-1249. https://doi.org/10.1001/jamaneurol.2022.3543

Langa, K. M., Weir, D. R., Kabeto, M., & Sonnega, A. (2023). Langa-Weir Classification of Cognitive Function (1995-2020). https://hrsdata.isr.umich.edu/sites/default/files/documentation/data-descriptions/1695907706/Data_Description_Langa_Weir_Classifications2020_V2.pdf

Hudomiet, P., Hurd, M. D., & Rohwedder, S. (2022). Trends in inequalities in the prevalence of dementia in the United States. Proceedings of the National Academy of Sciences, 119(46), e2212205119. https://doi.org/10.1073/pnas.2212205119

Table 3. Agreement among Core classifications and comparison diagnoses

Classification 1	Classification 2	Three-class quadratic weighted kappa	Dementia versus non-dementia kappa
Core algorithm	Manly-Jones algorithm	0.501	0.337
PMM	Manly-Jones algorithm	0.370	0.302
Langa-Weir	Manly-Jones algorithm	0.510	0.407
Hudomiet	Manly-Jones algorithm	0.551	0.457
HCAP Consensus	Manly-Jones algorithm	0.728	0.536
Core algorithm	Langa-Weir	0.558	0.406
PMM	Langa-Weir	0.425	0.348
Hudomiet	Langa-Weir	0.717	0.670
HCAP Consensus	Langa-Weir	0.099	0.736
Core algorithm	Hudomiet	0.601	0.442
PMM	Hudomiet	0.441	0.351
HCAP Consensus	Hudomiet	0.396	0.806
Core algorithm	HCAP Consensus	0.081	0.153
PMM	HCAP Consensus	0.138	0.249

Source: A7 rows: R_objects/A7_100_hcap_tables.rds, generated by R/A7_100-comparison_of_diagnoses.R. PMM rows: R/PMM_110_Comparison_of_Consensus_Langa_Weir.R; PMM-Hudomiet was calculated from the same saved probabilities and A7 comparison data. Three-class values are quadratic weighted kappas; binary values are kappas for dementia versus non-dementia. Consensus comparisons use normalized HCAP16WGTR / samplingP weights; other comparisons use HCAP16WGTR.

Table 4. Pairwise three-class agreement among HCAP comparison methods

method	Consensus diagnosis	Manly-Jones algorithm	Langa-Weir	Core algorithm	Hudomiet
Consensus diagnosis	1.000	0.728	0.099	0.081	0.396
Manly-Jones algorithm	0.728	1.000	0.510	0.501	0.551
Langa-Weir	0.099	0.510	1.000	0.558	0.717
Core algorithm	0.081	0.501	0.558	1.000	0.601
Hudomiet	0.396	0.551	0.717	0.601	1.000

Source: Source: R_objects/A7_100_hcap_tables.rds, generated by R/A7_100-comparison_of_diagnoses.R. Entries are quadratic weighted kappas. Consensus comparisons use normalized HCAP16WGTR / samplingP weights; other comparisons use HCAP16WGTR.

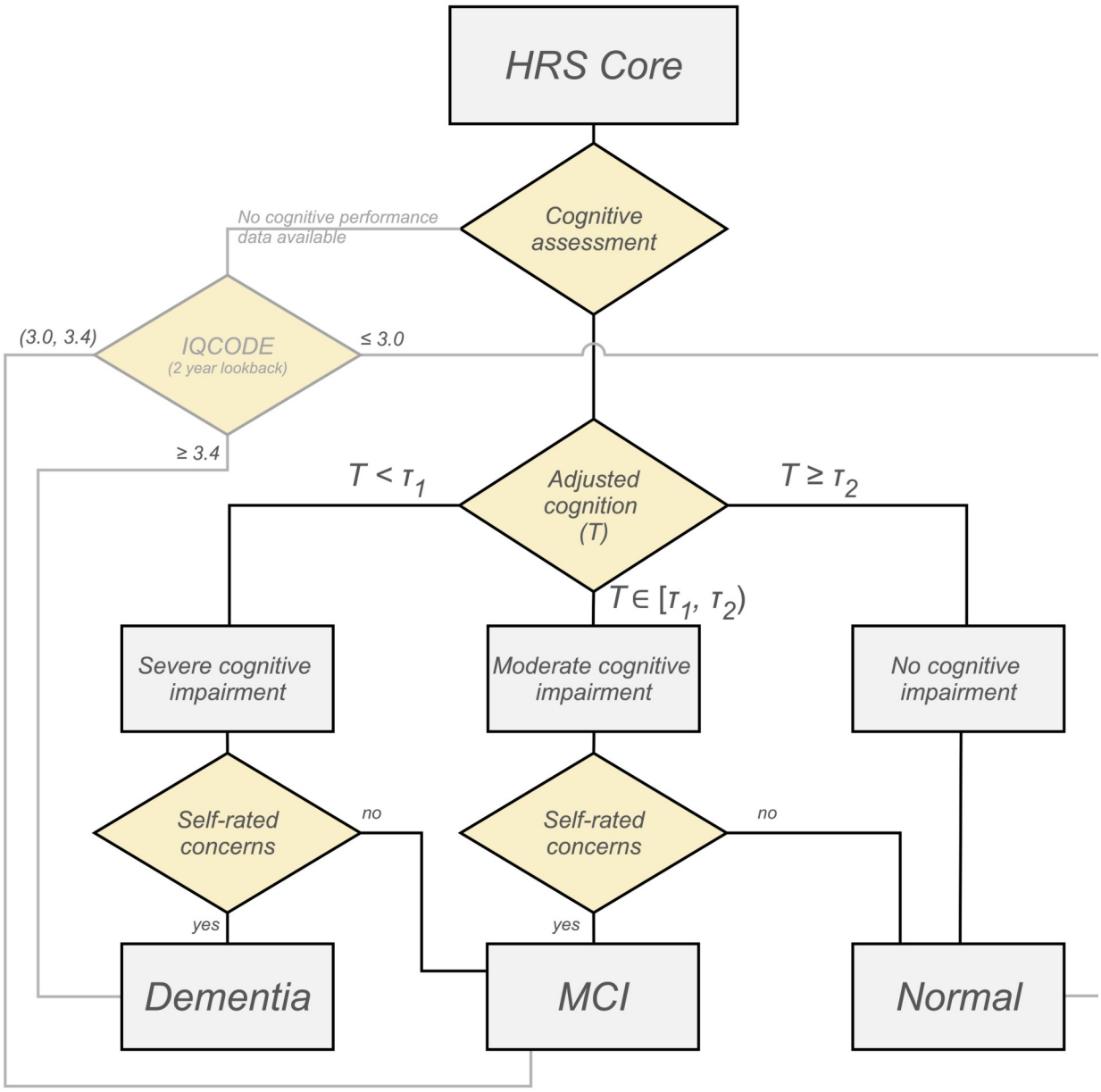


Figure 1: HRS core algorithm decision structure. T: demographically adjusted T score, rounded to an integer; τ_1 = 39 and τ_2 = 48: severe and moderate T-score cut points. IQCODE: Informant Questionnaire on Cognitive Decline in the Elderly (cut points 3.0 and 3.4). Self-rated concerns is a rating of memory worse than 2 years ago, and the "yes" arm includes don't know, refused, and not ascertained. Participant counts by branch are in Appendix 1.

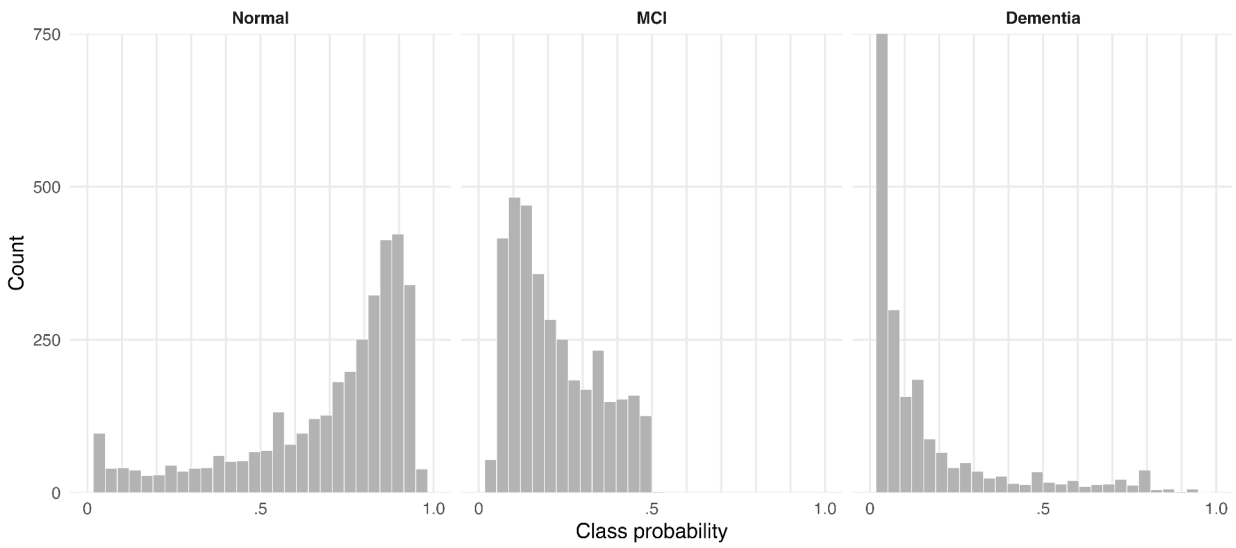


Figure 2: PMM class probabilities in HRS/HCAP

Thanks for reading.

Adaptation of the Manly-Jones algorithm to the HRS core survey data

2026-10-22

Introduction

The Manly et al. (2022) algorithm classifies participants as having normal cognition, mild cognitive impairment (MCI), or dementia from 5 cognitive performance domain scores estimated from an hour-long neuropsychological battery, together with an informant interview on cognitive and functional change, and a survey participant's self-reported memory problem (Manly et al., 2022). Of those indicators, only self-reported memory problems are available in the core interview. The HRS core interview includes a brief battery of scored cognitive items (8 used here), 1 self-rated memory item, and an informant questionnaire that is only administered when no cognitive performance data is available (Crimmins et al., 2011; Ofstedal et al., 2005). The core cognitive measures are too few to score separate domains defined in HCAP. The absence of informant reports on daily function when a respondent participates in the cognitive testing is an important additional difference in available information for cognitive classification.

To address these differences, we developed a new classification algorithm using the HRS core data, the "core algorithm", that follows the logic of the Manly-Jones (HCAP) algorithm but uses core data only. The core algorithm has 4 steps (Figure A1):

(1) a single cognitive performance score is estimated from the core items;

(2) the cognitive performance score is normed against the HRS-HCAP robust normative sample and expressed as a demographically adjusted T score;

(3) 2 integer cut points divide the T score, rounded to an integer, into 3 bands that stand in for the number of impaired domains; and

(4) decision rules combine the T-score band with self-rated memory change, or use the informant questionnaire when no cognitive tests were completed.

The core algorithm is defined using the HRS-HCAP 2016 sample. We use that sample's normative reference group to define participants' expected cognitive performance based on their age, sex, race, ethnicity, and years of completed education (Manly et al., 2022). We also use the HRS-HCAP 2016 sample and algorithm to set thresholds to calibrate the core model. Cognitive scoring statistical and psychometric procedures are adapted from Manly et al. (2022). Scoring parameters and norms were estimated once using core data in the HRS-HCAP Wave 1 sample. These estimates and the cut points were then held fixed and used for the classification

of participants in the 2016, 2018, 2020, and 2022 core interviews. A change in a participant's classification across waves will reflect changes in that participant's (or their informant's) responses and/or age, but the scoring model does not change. The same item thresholds, norms, and cut points convert a 2022 response pattern exactly as they would convert the same pattern observed in 2016.

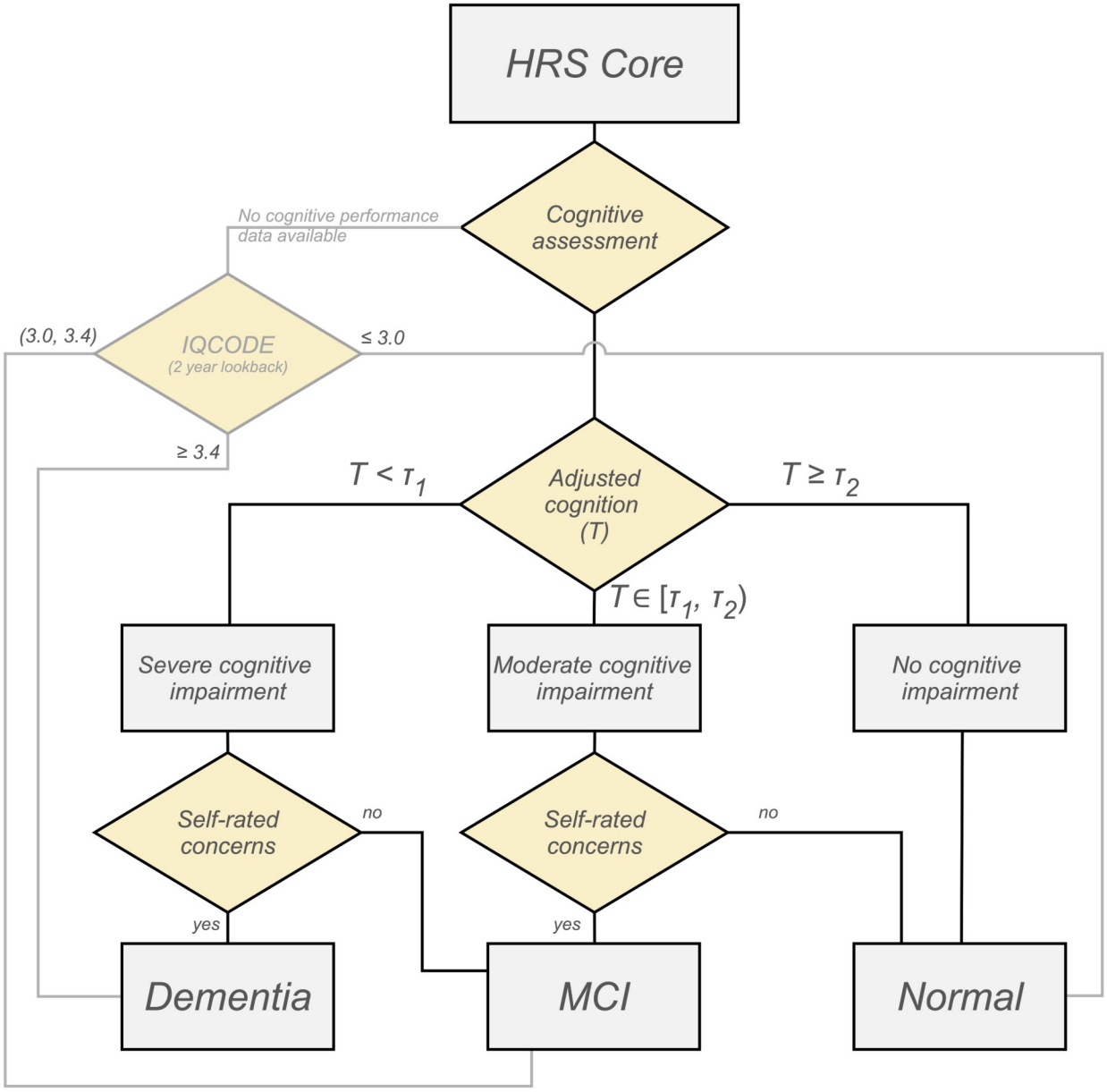


Figure A1. Core algorithm decision structure. T: demographically adjusted T score (Step 2); τ_1 and τ_2: severe and moderate T-score cut points (Step 3). IQCODE: Informant Questionnaire on Cognitive Decline in the Elderly. Self-rated concerns is a rating of memory worse than 2 years ago, and the "yes" arm includes don't know, refused, and not ascertained.

Data

The analytic sample is the 3,496 HRS-HCAP Wave 1 (2016) participants, followed in the HRS core interviews of 2016, 2018, 2020, and 2022. The core algorithm uses 4 kinds of core data:

(1) the 8 cognitive items in Table A1;

(2) self-rated memory compared with 2 years ago (HRS item D102: better, about the same, or worse);

(3) the 16-item informant questionnaire (IQCODE; Jorm, 1994), scored as the mean item rating from 1 (much improved) to 5 (much worse) and collected in proxy interviews in the core interview (2-year lookback); and

(4) age, sex, race and ethnicity (non-Hispanic Black, Hispanic), and years of schooling.

We judge the adequacy of the core algorithm by comparing it to the Manly et al. (2022) HRS-HCAP classification algorithm as obtained in the Wave 1 sample. Because the Manly et al. (2022) algorithm is used to calibrate the core algorithm, the agreement statistics we obtain are optimistic estimates of their agreement if the algorithms, as developed, were applied in an independent sample. We also compare to a separately derived HRS core classification algorithm referred to as the Langa-Weir classification, which was derived independent of HCAP data and methods (Crimmins et al., 2011).

Approach: Step 1. A single cognitive score from the core battery

The core cognitive performance data are used to generate a unidimensional model of general cognitive performance and generate a composite measure of functioning for participants. We begin this with a single-factor confirmatory factor analysis (CFA) model. This model uses a mix of binary, ordinal, and continuous indicators modeled as caused by a single general latent variable. The model was fit to HRS 2016 respondents aged 65 and older (N = 9,218) with the complex sample design and the HRS-HCAP Wave 1 weights. The use of HRS-HCAP Wave 1 weights means that those respondents not in the HRS-HCAP sample are included in the analysis with a weight of 0, so the estimates describe the population that HRS-HCAP represents (3,283 of the 3,496 HRS-HCAP participants with at least 1 HRS core item response). We evaluate model fit in the pairwise complete sample with the weighted least squares mean and variance adjusted (WLSMV) estimator fit using Mplus software (version 9, Muthén & Muthén, Los Angeles, CA). Then, item parameters were estimated by maximum likelihood with robust standard errors (MLR) and a probit link (for binary and ordinal items), which uses every

available item response regardless of item missingness. The WLSMV approach is consistent with the missing completely at random assumption, whereas the MLR approach is consistent with the less restrictive missing at random assumption. In subsequent factor score estimation runs, all parameters were fixed at the HCAP Wave 1 maximum likelihood estimates. Expected a posteriori (EAP) factor scores (the means of the posterior distribution of ability implied by the model parameters and assumptions about the distribution of ability; Bock & Mislevy, 1982) were estimated from a model with every parameter fixed and included all core person-wave respondents from 2016 to 2022 with at least 1 item response (63,992 person-waves, all ages).

Results: Step 1. A single cognitive score from the core battery

The single-factor model fit well (RMSEA = 0.035, 90% CI 0.031 to 0.039; CFI = 0.966; TLI = 0.952; SRMR = 0.047)[1] by the conventional guides. Standardized loadings in the MLR model ranged from 0.48 to 0.71 (Table A1), and that range indicates that no single item dominates the factor. As with our analysis of the HCAP battery, immediate word recall was not included in the model because it adds little information beyond delayed recall in the HCAP Wave 1 sample (Jones et al., 2024). Table A1 provides loadings from the MLR and WLSMV models. Loadings from the two estimators were similar for most items. The lower WLSMV loading for number series, the item with the most missing data (12% of HRS-HCAP participants with any core item response, against at most 1% for any other item), likely reflects how each estimator handles missing data. The higher WLSMV loading for object naming cannot be explained by missing data, which was nearly complete for that item. Instead, the difference may be related to the item's skewed response distribution (93% of HRS-HCAP participants named both objects correctly, weighted).

[1] RMSEA, root mean square error of approximation, estimates model misfit per degree of freedom in the population and does not compare the model with a baseline model. Lower values indicate better fit; values below 0.05 indicate close fit (Hu & Bentler, 1999; Steiger, 1990). CFI, comparative fit index, and TLI, Tucker-Lewis index, compare the fitted model with a baseline model in which the items are uncorrelated. Higher values indicate better fit; values of 0.95 or above indicate satisfactory fit (Bentler, 1990; Hu & Bentler, 1999; Tucker & Lewis, 1973). The TLI adds a penalty for model complexity. SRMR, standardized root mean square residual, is the average difference between the observed and model-implied correlations. Lower values indicate better fit; values of 0.08 or below indicate satisfactory fit (Asparouhov & Muthén, 2018; Hu & Bentler, 1999).

Table A1. HRS core cognitive items and standardized factor loadings

Item	Scoring	Loading (MLR, used for scoring)	Loading (WLSMV)
Orientation to date	Month, day, year, day of week; 0-4 correct (ordinal)	0.48	0.52
Animal naming	Animals named minus errors; rescaled to 0-1 (continuous)	0.60	0.63
Object naming (scissors, cactus)	0-2 correct (ordinal)	0.68	0.84
Naming the President and Vice President	0-2 correct (ordinal)	0.60	0.64
Delayed word recall	0-10 words; rescaled to 0-1 (continuous)	0.63	0.64
Number series	Score rescaled to 0-1 (continuous)	0.71	0.62
Serial 7 subtraction	0-5 correct (ordinal)	0.66	0.60
Backward counting from 20	Correct on first or second try (binary)	0.52	0.56

Single-factor model fit to HRS 2016 respondents aged 65 and older (N = 9,218) with HRS-HCAP weights (weight 0 outside HRS-HCAP). Loadings are standardized (STDYX). Continuous items were rescaled to 0-1 using the 2016 range. WLSMV fit: chi-square = 248.5 (df = 20), RMSEA = 0.035, CFI = 0.966, TLI = 0.952, SRMR = 0.047.

Approach: Step 2. Demographic norms

The Manly-Jones algorithm defines impairment as performance more than 1.5 standard deviations below the level expected for a person's age, sex, race, ethnicity, and education in a robust normative sample (Manly et al., 2022). Although we use different thresholds to define impairment (discussed below), we norm the single general cognitive domain score the same way we normed specific domain scores in Manly et al. (2022).

The HCAP 2016 robust normative sample (n = 1,787) is used to define expected cognitive performance levels. The robust normative sample is the subset of HRS-HCAP Wave 1 participants who can reasonably be assumed free of dementia or cognitive impairment, selected without reference to HCAP test scores as those who were followed up at the 2018 HRS core interview and had no reported or diagnosed stroke, Parkinson's disease, Alzheimer's disease or other dementia, cognitive impairment, or functional decline at HCAP baseline or follow-up. Their test performance defines the level expected for a person of a given age, sex, race, ethnicity, and education in the absence of a likely neurological condition (Manly et al., 2022).

First, factor scores in the normative sample were converted to normal scores with the Blom (1958) rank transformation. Because this transformation is sample dependent, we derive a linking equation between Blom transformed scores and estimated factor scores using linear regression with a restricted cubic spline for the factor score (knots at the 5th, 35th, 65th, and

95th percentiles; Harrell, 2015). The resulting regression model can be used to convert any factor score, including scores from people outside the normative sample and from later waves, to the normalized metric.

Second, the normalized score was regressed, with survey weights, on age (restricted cubic spline with knots at 70, 78, 86, 94 years, centered at 70), sex, non-Hispanic Black race, Hispanic ethnicity, and years of schooling (centered at 12), together with their 2-way interactions (24 terms). This demographic equation is used to estimate the expected score according to demographic characteristics.

Third, the sociodemographically adjusted cognitive performance score (T score) was computed as 50 + 10 × (observed − expected) / SEE, where SEE, the standard error of the estimate from the demographic regression in the normative sample, is the standard deviation of the normalized score multiplied by $\sqrt{(1 - R^2)}$. A T score of 40 therefore means performance 1 SEE below the level expected for a person of the same age, sex, race, ethnicity, and education. The same SEE is used regardless of demographic group, so a given T score marks the same distance below expectation for every participant.

Results: Step 2. Demographic norms

The demographic equation explained 37% of the variance in the normalized score in the normative sample. The standard deviation of the normalized score was 0.973, and therefore the SEE was 0.775 normalized-score units. In HRS-HCAP Wave 1, the unweighted mean (SD) T score was 50 (10) in the normative sample[2] and 43 (12) among participants excluded from the normative reference sample.

Approach: Step 3. Cut points

In the Manly-Jones algorithm, dementia requires at least 2 impaired domains and an informant report of functional impairment, and MCI requires at least 1 impaired domain with informant-reported concern or self-reported memory worsening (Manly et al., 2022).

No point on a single T score corresponds to 2 or more domains more than 1.5 standard deviations below expectation. With 5 domains, there are 10 different ways to satisfy exactly 2 domains impaired, and each would be associated with a different cut score on the single

[2] Manly et al. (2022) used the unweighted normative sample in this scale calibration step. An alternative would have been to use the sampling weights in the normative reference subpopulation group to set the mean (50) and standard deviation (10) on the rescaled cognitive performance measure. We use the convention established in Manly et al. (2022).

domain. We used 2 cut points on the T score (T representing the single domain cognitive performance measure, adjusted for sociodemographics and rescaled to a T distribution (M 50 SD 10) in the unweighted normative reference group): a severe cut point (τ_1) at rounded T < 39 and a moderate cut point (τ_2) at rounded T < 48. The 2 cut points stand in for the count of impaired domains: T below τ_1 corresponds to impairment in 2 or more domains, and T from τ_1 to below τ_2 corresponds to impairment in 1 domain.

We selected the 2 cut points with the following procedure: The T score was rounded to the nearest integer, and both cut points are integers; integer cut points are simple to apply and report, and a fractional cut point would imply a precision the T score does not have. We evaluated every integer pair of cut points (τ_1 from 25 to 50, τ_2 from 30 to 60, τ_2 above τ_1) in HRS-HCAP Wave 1, with the classification rules of Step 4 held fixed. A threshold pair was a candidate for selection if the weighted dementia prevalence of the core algorithm was within 0.5 percentage points of the Manly-Jones dementia prevalence and its weighted MCI prevalence was within 1.0 points of the Manly-Jones MCI prevalence. Among eligible pairs, we chose the pair with the highest weighted kappa for dementia versus no dementia, and broke ties with the weighted 3-class kappa (quadratic weights). If no pair had been identified as a candidate, the rule widened the dementia tolerance to 1.0 point (and no further) and then widened the MCI tolerance in steps of 0.5 points. The rule puts prevalence first because the core algorithm is used to estimate prevalence and incidence in waves without an HCAP assessment, and puts dementia agreement second because several pairs can match prevalence equally well.

Results: Step 3. Cut points

In HRS-HCAP Wave 1 (n = 3,496), 5 of 575 pairs of thresholds were identified as candidates. The rule selected $\tau_1 = 39$ and $\tau_2 = 48$, so a rounded T score below 39 marks the severe range and a rounded T score from 39 to 47 marks the moderate range of the T score. At these cut points the weighted dementia kappa was 0.34 and the 3-class kappa was 0.50; Table A3 gives the class prevalence estimates. To describe how firm the selection is, we repeated the whole rule in 500 bootstrap samples drawn by the HRS sampling design; the rule selected the same pair in 8% of the bootstrap samples and a pair within 1 point of both cut points in 50% of the bootstrap samples.

The same pair recurs in few bootstrap samples because the rule compares prevalence in whole T-score steps against limits of 0.5 and 1.0 percentage points. Next to the selected cut points, a 1-point change in τ_1 changed the weighted dementia prevalence by 0.5 percentage points, and

a 1-point change in τ_2 changed the weighted MCI prevalence by 0.9 points. In a new sample, the prevalence from both algorithms shifts by a fraction of a point, and that shift can move a pair into or out of the candidate set. The rule fixes one pair, but the data do not separate it from its neighbors: pairs within 1 point of both cut points were selected in 50% of the bootstrap samples.

Figure A2 shows the weighted distribution of T scores among HRS-HCAP Wave 1 participants with core cognitive tests, with the 2 cut points and the weighted share of participants in each T-score band.

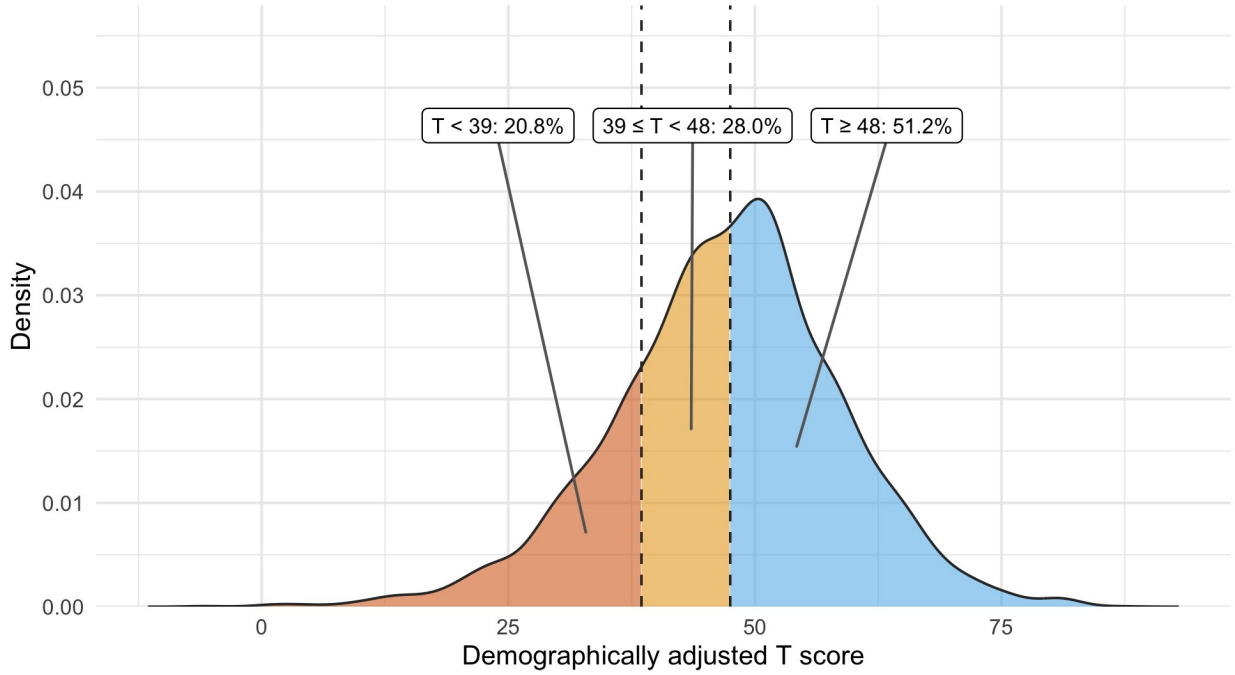


Figure A2. Weighted distribution of demographically adjusted T scores among HRS-HCAP Wave 1 participants with core cognitive tests (n = 3,283), with the cut points (dashed lines). Cut points apply to the T score rounded to an integer, so each dashed line sits 0.5 below its cut point. Band labels give the weighted share of these participants in each band of the rounded T score.

Step 4. Classification rules

The core interview has no informant report for respondents who participated in the cognitive assessment (Figure A1; Table A2). The core interview asks every self-respondent about limitations in activities of daily living (ADL) and instrumental activities of daily living (IADL), and these self-reports could stand in for the informant report of functional impairment that separates dementia from MCI in the Manly-Jones algorithm. We did not use them because the level and kind of impairments captured by the core interview ADL and IADL limitations are fundamentally different from those captured by informant IQCODE and Blessed Part I reports in HCAP. Few participants report any limitation, even among those the Manly-Jones algorithm classifies as

having dementia. Among HRS-HCAP Wave 1 participants with a core T score and dementia by the Manly-Jones algorithm, 52.3% (weighted) reported no ADL or IADL limitation. The Appendix Endnote addresses this point more fully.

Self-rated memory worsening, an item the Manly-Jones algorithm already includes, is the second criterion at both levels and stands in for the informant report. Participants without core cognitive tests (mostly proxy interviews) were classified from the IQCODE with the Manly-Jones informant thresholds (3.4 or higher for dementia; above 3.0 for MCI). Missing self-rated memory was singly imputed by chained equations (van Buuren & Groothuis-Oudshoorn, 2011); no Wave 1 participant with a T score was missing self-rated memory, and “don’t know,” refused, and not ascertained responses (4 participants) were counted as “worse”, which assigns these participants to the more impaired class.

Table A2. Core algorithm decision rules and participants classified by each rule, HRS-HCAP Wave 1

Route	Cognitive criterion	Second criterion	Classification	n
Cognitive tests	T < 39	Memory worse than 2 years ago*	Dementia	237
Cognitive tests	T < 39	Memory better or the same	MCI	487
Cognitive tests	39 ≤ T < 48	Memory worse than 2 years ago*	MCI	270
Cognitive tests	39 ≤ T < 48	Memory better or the same	Normal	674
Cognitive tests	T ≥ 48	Any	Normal	1,615
Proxy interview	No T score	IQCODE ≥ 3.4	Dementia	107
Proxy interview	No T score	3.0 < IQCODE < 3.4	MCI	37
Proxy interview	No T score	IQCODE ≤ 3.0	Normal	69

n: unweighted HRS-HCAP Wave 1 participants (N = 3,496; Normal 2,358, MCI 794, Dementia 344). T: demographically adjusted T score (mean 50, SD 10 in the HRS-HCAP robust normative sample). *Self-rated memory compared with 2 years ago (HRS item D102); don't know, refused, and not ascertained count as worse; missing values were singly imputed. IQCODE: 16-item Informant Questionnaire on Cognitive Decline in the Elderly, mean item score (1-5), asked in proxy interviews.

With these rules, the weighted prevalence estimates of the core algorithm in HRS-HCAP Wave 1 were 68.9% normal, 22.0% MCI, and 9.1% dementia, compared with 69.4%, 21.4%, and 9.2% for the Manly-Jones algorithm estimated with EAP factor scores (Table A3).

Table A3. Weighted class prevalence of the core algorithm and the Manly-Jones algorithm, HRS-HCAP Wave 1

Classification	Normal, %	MCI, %	Dementia, %
Core algorithm	68.9	22.0	9.1
Manly-Jones algorithm	69.4	21.4	9.2

HRS-HCAP Wave 1, N = 3,496. Percentages are weighted (HCAP Wave 1 (2016) sampling weight). Core algorithm cut points:

Table A3. Weighted class prevalence of the core algorithm and the Manly-Jones algorithm, HRS-HCAP Wave 1			
Classification	Normal, %	MCI, %	Dementia, %

rounded T < 39 (severe) and rounded T < 48 (moderate).

Agreement with the Manly-Jones algorithm in HRS-HCAP Wave 1

We next examine the agreement of the core algorithm with the Manly-Jones classification and compare it with the agreement of the Langa-Weir classification, a widely used classification of cognitive status from HRS core data (Crimmins et al., 2011). For self-respondents, Langa-Weir sums immediate and delayed word recall (0 to 20), serial 7 subtraction (0 to 5), and backward counting (0 to 2) into a 27-point score, and classifies scores of 0 to 6 as dementia, 7 to 11 as cognitive impairment without dementia (CIND), and 12 to 27 as normal. For proxy interviews, it combines the proxy's report of the respondent's memory, limitations in instrumental activities of daily living, and the interviewer's assessment of cognitive impairment into an 11-point score, with 6 or more classified as dementia and 3 to 5 as CIND (cognitive impairment, not demented; Crimmins et al., 2011, pp. i164, i166). The Langa-Weir cut points were chosen so that the HRS distribution of cognitive states matches the distribution estimated in the Aging, Demographics, and Memory Study (ADAMS). The Langa-Weir score is not adjusted for demographic characteristics. We treat CIND as equivalent to MCI for this comparison.

Agreement of the core algorithm with the Manly-Jones algorithm for 3-class Normal/MCI/Dementia classification was about the same as the agreement of Langa-Weir; for dementia versus not, it was lower than that of Langa-Weir (Table A4). The quadratic weighted kappa (Cohen, 1968) for the 3 classes was 0.50 for the core algorithm and 0.51 for Langa-Weir; for dementia versus not, kappa was 0.34 and 0.41. Of the weighted population the Manly-Jones algorithm classified as having dementia, the core algorithm classified 40% as having dementia and Langa-Weir classified 42%. Of those the core algorithm classified as having dementia, 40% had dementia by the Manly-Jones algorithm (Langa-Weir: 50%). Agreement between the core algorithm and Langa-Weir was 0.56 (3 classes, weighted kappa) and 0.41 (dementia, kappa). Table A5 gives the weighted cross-classifications. Each kappa compares 2 classifications that both contain error, so these values describe agreement and not accuracy.

Table A4. Agreement of the core algorithm and Langa-Weir with the Manly-Jones algorithm, HRS-HCAP Wave 1

Comparison	Kappa, 3-class (quadratic)	Kappa, dementia vs. not	Manly-Jones dementia classified as dementia, %	Comparator dementia that is Manly-Jones dementia, %
Core algorithm vs. Manly-Jones	0.50	0.34	40	40
Langa-Weir vs. Manly-Jones	0.51	0.41	42	50
Core algorithm vs. Langa-Weir	0.56	0.41	n/a	n/a

N = 3,496 HRS-HCAP Wave 1 participants with all 3 classifications; statistics weighted with HCAP Wave 1 (2016) sampling weight. Core algorithm and Langa-Weir use 2016 HRS core data; the Manly-Jones algorithm uses HRS-HCAP data. Kappa, 3-class: Normal, MCI, Dementia with quadratic weights (Cohen, 1968). The last 2 columns treat the Manly-Jones classification as the reference; n/a, not applicable.

Table A5. Weighted cross-classification of the core algorithm and Langa-Weir with the Manly-Jones algorithm, HRS-HCAP Wave 1

Comparator	Manly-Jones class	Comparator class: Normal	Comparator class: MCI	Comparator class: Dementia	Total
Core algorithm	Normal	28,638 (82%)	5,145 (15%)	1,005 (3%)	34,788
Core algorithm	MCI	5,098 (48%)	3,876 (36%)	1,751 (16%)	10,725
Core algorithm	Dementia	794 (17%)	1,995 (43%)	1,831 (40%)	4,620
Langa-Weir	Normal	30,246 (87%)	4,091 (12%)	450 (1%)	34,788
Langa-Weir	MCI	6,070 (57%)	3,120 (29%)	1,535 (14%)	10,725
Langa-Weir	Dementia	1,220 (26%)	1,439 (31%)	1,961 (42%)	4,620

Rows: Manly-Jones class (HRS-HCAP). Columns: comparator class (core algorithm or Langa-Weir) from 2016 HRS core data. Cell entries are survey-weighted population counts in thousands. Entries for fewer than 1,000 persons show two significant digits. Row percentages in parentheses. N = 3,496 participants with both classifications (HCAP Wave 1 (2016) sampling weight). Kappas are in Table A4.

Application to the 2018, 2020, and 2022 core interviews

The fixed model, norms, and cut points were applied to every core interview of HRS-HCAP Wave 1 participants from 2016 to 2022 (Table A6). The number of classified person-waves fell from 3,496 in 2016 to 2,049 in 2022 as participants died or left the study, and the share classified from the IQCODE alone was 6.1% in 2016 and 6.9% in 2022.

Table A6. Core algorithm classifications of HRS-HCAP Wave 1 participants interviewed in the HRS core, 2016-2022

Core wave	Person-waves	Cognitive tests	IQCODE only	Not classified	Normal	MCI	Dementia
2016	3,496	3,283	213	0	2,358 (67.4%)	794 (22.7%)	344 (9.8%)
2018	2,989	2,836	152	1	2,110 (70.6%)	631 (21.1%)	247 (8.3%)
2020	2,558	2,380	178	0	1,879 (73.5%)	478 (18.7%)	201 (7.9%)
2022	2,050	1,907	142	1	1,449 (70.7%)	399 (19.5%)	201 (9.8%)

Unweighted counts of person-waves; percentages among classified person-waves. Not classified: no cognitive tests and no IQCODE. Number series was not administered in the 2018 core interview, so 2018 T scores rest on 7 items.

Missing Number Series

Number series was not administered in the 2018 core interview (no 2018 respondent has a score), so 2018 EAP scores are based on 7 items. With less information, an EAP score is pulled further toward the population mean (Bock & Mislevy, 1982), which raises the scores of low performers and may classify fewer participants as impaired in 2018 than the same performance would produce in other waves. We did not re-norm the 7-item score because the 7 items are in common with the 2016 core items and factor scores are estimated with fixed parameter estimates, therefore the factor score estimates are on the same scale (fixed parameter calibration).

Figure A3 and Table A7 compare EAP scores of core respondents aged 65 or older in 2018 (when number series was not administered) with those in 2016 and 2020 (combined, when number series was administered). The mean posterior standard deviation of the EAP score (the uncertainty in one participant's score given that participant's item responses, averaged over participants) was 0.56 in 2018 and 0.49 in 2016 and 2020, and the standard deviation of the EAP scores (the spread of the score estimates across participants) was 0.85 and 0.89. The lower quartile was -0.60 in 2018 and -0.65 in 2016 and 2020. The effect on the spread of scores was small; the standard deviation of the 2018 EAP scores was about 4% smaller than in 2016 and 2020 (Table A7). Nevertheless, a small shift for the population can still change the classification of individuals whose scores lie near a cut point. Because shrinkage moves low scores toward the mean, a participant whose T score would fall just below τ_1 with number series may fall just above it without it, and be classified with MCI rather than dementia.

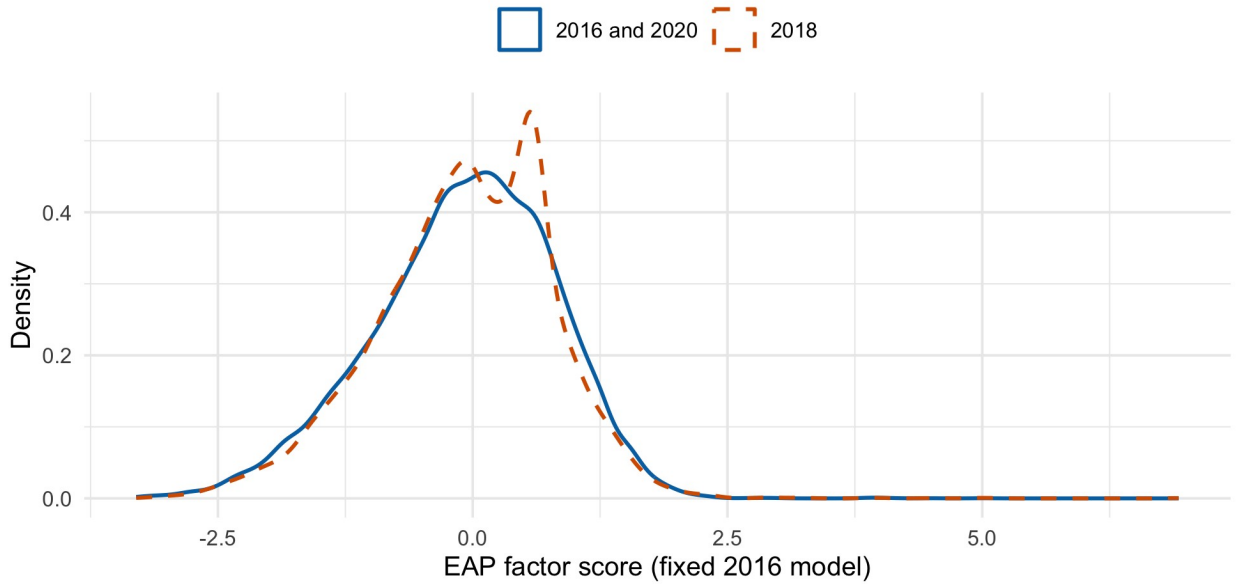


Figure A3. Distribution of EAP factor scores among core respondents aged 65 or older with a nonzero wave weight, in waves with number series (2016 and 2020) and without it (2018). Scores come from the fixed 2016 model.

Table A7. EAP factor scores in core waves with and without number series

Core waves	Person-waves	Minimum	Q1	Median	Q3	Maximum	SD of EAP	Mean posterior SD	Var(EAP) + mean posterior variance
2016 and 2020	17,518	-3.30	-0.65	-0.02	0.55	5.01	0.89	0.49	1.03
2018	8,404	-3.26	-0.60	-0.03	0.59	6.93	0.85	0.56	1.05

Core respondents aged 65 or older with a nonzero wave weight and at least 1 cognitive item; unweighted. 2016 and 2020: number series administered; 2018: not administered. 2022 is omitted because number series is missing for most face-to-face interviews. Under the model, the variance of the EAP scores plus the mean posterior variance equals the factor variance (1), so a larger posterior SD implies a smaller spread of EAP scores (shrinkage toward the mean).

Table A8 lists the number of core respondents aged 65 or older with a score on each test, by wave. Number series was present for 84% (weighted) of respondents in 2016, 86% in 2020, and 49% in 2022, and absent in 2018; the other tests were present for at least 78% of respondents in every wave.

Table A8. Core cognitive tests with a score, by wave: number (weighted %)

Test	2016	2018	2020	2022
Orientation to date	9,158 (94.9%)	7,578 (83.8%)	7,219 (78.3%)	6,751 (79.4%)
Animal naming	9,092 (94.2%)	7,404 (82.0%)	7,218 (78.3%)	6,716 (79.0%)
Object naming	9,158 (94.9%)	7,578 (83.8%)	7,219 (78.3%)	6,751 (79.4%)
President and Vice President	9,157 (94.8%)	7,578 (83.8%)	7,219 (78.3%)	6,751 (79.4%)
Delayed word recall	9,094 (94.1%)	7,493 (82.9%)	7,157 (77.5%)	6,641 (78.1%)
Number series	7,875 (84.2%)	0 (0.0%)	7,274 (85.8%)	3,897 (49.4%)
Serial 7 subtraction	9,158 (94.9%)	8,404 (95.7%)	8,360 (94.8%)	7,759 (95.4%)
Backward counting from 20	9,121 (94.5%)	7,535 (83.4%)	7,194 (77.9%)	6,725 (79.1%)
Immediate word recall (not in model)	9,095 (94.2%)	7,514 (83.1%)	7,185 (77.8%)	6,691 (78.7%)
Any cognitive item	9,158 (94.9%)	8,404 (95.7%)	8,383 (95.1%)	7,759 (95.4%)

Core respondents aged 65 or older with a nonzero wave weight (2016: n = 9,684; 2018: n = 8,822; 2020: n = 8,869; 2022: n = 8,161). Counts are unweighted; percentages are weighted with the wave respondent weight. Proxy interviews, which have no cognitive tests, are included in the denominator.

Limitations

Limitations of our approach are worth mentioning.

First, the cut points were selected in the HRS-HCAP Wave 1 sample, in which agreement is also evaluated, so the agreement in Table A4 is optimistic. The comparison of core-based and HRS-HCAP-based classifications among Wave 2 re-interviewees (eTables 7 and 8) is the out-of-sample check available in these data.

Second, self-rated memory replaces informant report, and most participants with low T scores did not report memory worsening; the core algorithm likely under-classifies dementia relative to the Manly-Jones algorithm, which is consistent with the lower core-based dementia incidence among re-interviewees in eTable 8.

Third, the 2018 scores rest on 7 items (above). The pattern of missingness for number series is difficult to discern from the data and the HRS concordance web tool.

Fourth, the norms do not account for interview mode, which varies across respondents and waves; a mode difference in test scores would shift T scores in the waves where the mode mix differs from 2016. Other potentially important considerations, such as language of administration, are not incorporated into the scoring and standardizing steps.

Fifth, the IQCODE in the core interview asks about change over 2 years, whereas the HRS-HCAP informant interview asks about change over 10 years, so the Manly-Jones IQCODE thresholds may not carry the same meaning in core proxy interviews.

Finally, both the core algorithm and the Manly-Jones algorithm rely on applying strict cut points to continuous cognitive, functional, and informant report measures. This practice of attempting to elicit qualitative clinical states from quantitative scales introduces well-documented classification instability, particularly for individuals whose scores fall near diagnostic thresholds (Altman & Royston, 2006; Streiner, 2002). Because continuous cognitive instruments inherently possess standard measurement error, small and clinically negligible fluctuations in test performance can shift an individual across an arbitrary boundary. As a result, observations that are quantitatively almost indistinguishable are operationalized as qualitatively distinct diagnostic categories, substantially inflating misclassification rates and generating spurious diagnostic changes across repeated assessments.

Endnote. Why the core algorithm does not use functional limitations

In the Manly-Jones algorithm, an informant report of functional impairment separates dementia from MCI (Manly et al., 2022), and the core interview asks every self-respondent about limitations in activities of daily living (ADL) and instrumental activities of daily living (IADL). We planned to use these self-reports in place of the informant report and to set their thresholds by prevalence matching within each level of cognitive impairment. We set the plan aside for 3 reasons, which we document here with the current data. The first and third reasons are summarized in Step 4. Limitations were counted on the 9 ADL and IADL items asked in every core wave from 2016 to 2022 (Table A9 note), so that the count would mean the same thing in every wave.

Few participants report any limitation, even among those the Manly-Jones algorithm classifies as having dementia. Among HRS-HCAP Wave 1 participants with a core T score, the weighted share reporting no ADL or IADL limitation was 52.3% for those with dementia by the Manly-Jones algorithm, 68.7% for MCI, and 84.0% for normal cognition; below the severe T cut point, 66.1% reported none (Table A9). By definition, every participant the Manly-Jones algorithm classifies as having dementia has an informant report of functional impairment, yet 52.3% of them report no limitation in the core interview. The lowest possible threshold, 1 or more limitations, misses at least that share, and no threshold on the limitation count can do better.

Including function raised agreement by a small amount, and never enough to justify the apparent difference in measured construct. We examined 2 alternative rules, all at the core algorithm's cut points (T < 39 and T < 48), so that the rules differ only in the second criterion: function at the severe level and function or memory at the moderate level, the closest analog of the Manly-Jones rules; and function or memory at both levels (Table A10). In the full sample, the 3-class kappa was 0.50 with the memory rule, 0.53 with the function rule, and 0.52 with function or memory; the dementia kappa was 0.34, 0.39, and 0.38. Because the cut points are held fixed, the rules also produce different class prevalence (Table A10). To describe how much these differences vary across samples, we computed agreement in each of 200 random halves of the sample (Table A11). Relative to the memory rule, the mean difference in 3-class kappa across halves was +0.02 (+0.01 to +0.04) for the function rule and +0.02 (0.00 to +0.03) for function or memory (2.5th to 97.5th percentile across halves), and the function rule had the higher kappa in 98% of halves. For dementia, the mean differences were +0.05 (0.00 to +0.11) and +0.05 (+0.01 to +0.08). The gain is therefore small but consistent across halves, and our decision rests on the third reason.

The core items do not say why a limitation is present and do not specifically ask about change the way the IQCODE and Blessed Part I measures do in HRS-HCAP, and they rely on self-report in the core, whereas in HRS-HCAP a knowledgeable informant provides the assessment of change (unspecified time frame in the Blessed, a 10-year lookback in the IQCODE).

Table A9. Self-reported ADL and IADL limitations by Manly-Jones class and by core T-score band, HRS-HCAP Wave 1

Grouping	Group	n	None, %	1, %	2 or more, %	Missing, %
Manly-Jones class	Normal	2,256	84.0	8.8	7.2	0.0
Manly-Jones class	MCI	744	68.7	14.2	17.1	0.0
Manly-Jones class	Dementia	283	52.3	14.1	33.6	0.0
Core T-score band	T < 39	724	66.1	12.5	21.4	0.0
Core T-score band	39 to < 48	944	78.8	9.2	12.1	0.0
Core T-score band	T >= 48	1,615	83.5	10.1	6.4	0.0

Participants with a core T score (n = 3,283). Number of limitations in activities of daily living (ADL) and instrumental activities of daily living (IADL) reported in the 2016 core interview, counted on the 9 items asked in every wave from 2016 to 2022 (dressing, bathing, eating, and using the toilet; preparing meals, shopping for groceries, using the telephone, taking medications, and managing money). Missing: no response to any of these items. Percentages are weighted (HCAP Wave 1 (2016) sampling weight).

Table A10. Agreement with the Manly-Jones algorithm when functional limitations enter the decision rules, HRS-HCAP Wave 1

Rule for the second criterion	Severe cut	Moderate cut	MCI, %	Dementia, %	Kappa, 3-class	Kappa, dementia	Manly-Jones dementia classified as dementia, %
Memory worse (core algorithm)	39	48	22.0	9.1	0.50	0.34	40
Function at severe level; function or memory at moderate level	39	48	24.7	9.7	0.53	0.39	46
Function or memory at both levels	39	48	21.3	13.0	0.52	0.38	54

N = 3,496. All rules use the core algorithm's cut points (T < 39 and T < 48), so the rules differ in class prevalence as well as agreement. Function: 1 or more ADL or IADL limitations on the items in Table A9 (missing counted as none). Memory: memory worse than 2 years ago. Statistics are weighted (HCAP Wave 1 (2016) sampling weight); kappa, 3-class uses quadratic weights. Table A11 shows how these statistics vary across random halves of the sample.

Table A11. Agreement with the Manly-Jones algorithm in repeated random halves, HRS-HCAP Wave 1

Rule for the second criterion	Kappa, 3-class	Difference from memory rule	Halves higher, %	Kappa, dementia	Difference from memory rule	Halves higher, %
Memory worse (core algorithm)	0.50 (0.46, 0.54)	n/a	n/a	0.34 (0.27, 0.40)	n/a	n/a
Function at severe level; function or memory at moderate level	0.53 (0.49, 0.56)	+0.02 (+0.01, +0.04)	98	0.39 (0.33, 0.44)	+0.05 (0.00, +0.11)	97
Function or memory at both levels	0.52 (0.48, 0.56)	+0.02 (0.00, +0.03)	99	0.38 (0.33, 0.44)	+0.05 (+0.01, +0.08)	99

200 random halves of the sample, drawn within Manly-Jones class. In each half, agreement was computed for each rule at the core algorithm's cut points (T < 39 and T < 48). Entries are the mean across halves (2.5th and 97.5th percentiles). Differences are paired within half. The halves overlap, so the percentiles describe variation across halves and are not confidence intervals. Statistics are weighted (HCAP Wave 1 (2016) sampling weight); kappa, 3-class uses quadratic weights.

References

Altman, D. G., & Royston, P. (2006). The cost of dichotomising continuous variables. *BMJ*, *332*(7549), 1080. https://doi.org/10.1136/bmj.332.7549.1080

Asparouhov, T., & Muthén, B. (2018). *SRMR in Mplus*. Technical note, Muthén & Muthén. https://www.statmodel.com/download/SRMR2.pdf

Bentler, P. M. (1990). Comparative fit indexes in structural models. *Psychological Bulletin*, *107*(2), 238–246. https://doi.org/10.1037/0033-2909.107.2.238

Blom, G. (1958). *Statistical estimates and transformed beta-variables*. Wiley.

Bock, R. D., & Mislevy, R. J. (1982). Adaptive EAP estimation of ability in a microcomputer environment. *Applied Psychological Measurement*, *6*(4), 431–444. https://doi.org/10.1177/014662168200600405

Cohen, J. (1968). Weighted kappa: Nominal scale agreement with provision for scaled disagreement or partial credit. *Psychological Bulletin*, *70*(4), 213–220. https://doi.org/10.1037/h0026256

Crimmins, E. M., Kim, J. K., Langa, K. M., & Weir, D. R. (2011). Assessment of cognition using surveys and neuropsychological assessment: The Health and Retirement Study and the Aging, Demographics, and Memory Study. *The Journals of Gerontology, Series B: Psychological Sciences and Social Sciences*, *66B*(S1), i162–i171. https://doi.org/10.1093/geronb/gbr048

Harrell, F. E., Jr. (2015). *Regression modeling strategies* (2nd ed.). Springer.

Hu, L., & Bentler, P. M. (1999). Cutoff criteria for fit indexes in covariance structure analysis: Conventional criteria versus new alternatives. *Structural Equation Modeling*, *6*(1), 1–55. https://doi.org/10.1080/10705519909540118

Jones, R. N., Manly, J. J., Langa, K. M., Ryan, L. H., Levine, D. A., McCammon, R., & Weir, D. (2024). Factor structure of the Harmonized Cognitive Assessment Protocol neuropsychological battery in the Health and Retirement Study. *Journal of the International Neuropsychological Society*, *30*(1), 47–55. https://doi.org/10.1017/S135561772300019X

Jorm, A. F. (1994). A short form of the Informant Questionnaire on Cognitive Decline in the Elderly (IQCODE): Development and cross-validation. *Psychological Medicine*, *24*(1), 145–153. https://doi.org/10.1017/S003329170002691X

Manly, J. J., Jones, R. N., Langa, K. M., Ryan, L. H., Levine, D. A., McCammon, R., Heeringa, S. G., & Weir, D. (2022). Estimating the prevalence of dementia and mild cognitive impairment in the US: The 2016 Health and Retirement Study Harmonized Cognitive Assessment Protocol Project. *JAMA Neurology*, *79*(12), 1242–1249. https://doi.org/10.1001/jamaneurol.2022.3543

Ofstedal, M. B., Fisher, G. G., & Herzog, A. R. (2005). *Documentation of cognitive functioning measures in the Health and Retirement Study* (HRS Documentation Report DR-006). University of Michigan. https://doi.org/10.7302/24800

Steiger, J. H. (1990). Structural model evaluation and modification: An interval estimation approach. *Multivariate Behavioral Research*, *25*(2), 173–180. https://doi.org/10.1207/s15327906mbr2502_4

Streiner, D. L. (2002). Breaking up is hard to do: The heartbreak of dichotomizing continuous data. *The Canadian Journal of Psychiatry*, *47*(3), 262–266. https://doi.org/10.1177/070674370204700307

Tucker, L. R., & Lewis, C. (1973). A reliability coefficient for maximum likelihood factor analysis. *Psychometrika*, *38*(1), 1–10. https://doi.org/10.1007/BF02291170

van Buuren, S., & Groothuis-Oudshoorn, K. (2011). mice: Multivariate imputation by chained equations in R. *Journal of Statistical Software*, *45*(3), 1–67. https://doi.org/10.18637/jss.v045.i03

Appendix 2. PMM and logistic regression

2026-10-22

PMM and logistic regression can both classify people into Normal, MCI, or Dementia. The difference is what they model.

Logistic regression

Logistic regression starts with a known outcome, Y, and predicts it from observed predictors, X:

$$P(Y = \text{Dementia} \mid X)$$

For example, it could predict dementia from word recall, serial sevens, self-rated memory, and Jorm score. It estimates how each predictor changes the log-odds of each outcome category. In a multinomial logistic model, the usual form is:

$$\log \frac{P(Y = k \mid X)}{P(Y = \text{Normal} \mid X)} = \beta_{0k} + \beta_{1k} X_1 + \cdots + \beta_{pk} X_p$$

The main modeling decision is the regression equation: which predictors, interactions, nonlinear terms, and coefficients to include.

PMM in this project

The PMM takes a generative approach. It asks: *If a person belongs to the Normal, MCI, or Dementia class, what pattern of cognitive-test, self-report, and informant responses would we expect?* PMM models:

$$P(X \mid C = k)$$

for each cognitive class $C = k$, along with the class prevalence $P(C = k)$. It then uses Bayes' rule to obtain the classification probability:

$$P(C = k \mid X) = \frac{P(X \mid C = k) P(C = k)}{\sum_j P(X \mid C = j) P(C = j)}$$

So, instead of directly predicting diagnosis from scores, it describes the characteristic response distribution within each diagnosis class and asks which class best accounts for a person's observed pattern.

For the final PMM, the cognition model uses a general cognitive factor measured by the 8 individual test scores, plus the self-rated memory item. The classes differ in the mean and variance of the factor, and the factor carries the correlation among the tests within each class. The model estimates class probabilities from that whole response pattern. The separate Jorm PMM does the analogous job for people with informant data rather than cognitive testing.

Practical distinctions

- Logistic regression: “Given these scores, what is the probability of Dementia?”
- PMM: “Which diagnosis class most plausibly generated this pattern of scores and reports?”
- Logistic regression is discriminative: it models $P(Y \mid X)$ directly.
- PMM is generative: it models $P(X \mid C)$, then derives $P(C \mid X)$.
- Logistic regression commonly treats predictors as explanatory variables with regression coefficients.
- PMM treats the observed responses as indicators of an underlying class-specific response profile.
- Logistic regression needs explicit interactions or nonlinear terms when the meaning of one score depends on another.
- A PMM represents class differences through class-specific response distributions. In this project those differences act through the general cognitive factor, which also carries the dependence among the tests within each class.

In this project, the important wrinkle is that the PMM is not purely exploratory latent class analysis. It is calibrated using the Manly-Jones (HCAP) algorithm classification as the known class in the HCAP sample, then scores the Core measures of the same HCAP participants with the calibrated parameters (including the class proportions) held fixed. That makes it a supervised, model-based classifier: conceptually closer to generative discriminant analysis than to an unsupervised profiling exercise.

A concise manuscript description would be:

> *The PMM is a supervised, generative classifier that estimates the probability of each Manly-Jones algorithm class from the joint pattern of observed cognitive, self-report, or informant responses.*

Appendix 3: Comparisons with other study classifications

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